

B.TECH COMPUTER SCIENCE AND ENGINEERING

Team Vision X4

Academic Project Proposal

INSTITUTION	R.D. Engineering College
PROJECT GUIDE	Mr. Vivek
PROPOSAL FOCUS	Improving construction-material business operations

A proposed digital solution for clearer records, faster service and more connected operations.

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WHAT · THE CURRENT FRICTION

Traditional operations create avoidable friction

Construction-material businesses rely on manual records, phone calls and scattered conversations, slowing service and reducing reliability for customers and owners.



Manual orders

Requests are hard to record, update and verify.



Scattered credit

Udhaar records span notebooks, messages and memory.



Unclear payments

Cash, UPI and pending balances are hard to reconcile.



Repeated queries

Product details and order status are not visible in one place, so owners repeat answers.



No digital catalogue

Customers cannot compare products, quantities and basic information in one place.

THE PATTERN

One request, many places

Notebook → phone call → payment note → follow-up question

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WHY · THE VALUE OF DIGITISATION

Digitisation improves clarity, speed and trust

A proposed digital system can bring product information, orders, payments and customer records into one connected workflow. The value is practical: fewer repeated tasks, clearer communication and better visibility.



For customers

Browse products, place orders, ask questions, view payment options and find the business location easily.



For business owners

Manage orders, review customer balances, track payments and monitor activity from one dashboard.



For the business

Replace scattered information with consistent, searchable digital records.

Traditional system and proposed digital system

AREA	TRADITIONAL SYSTEM	PROPOSED DIGITAL SYSTEM
Product information	Phone calls and physical references	Digital catalogue
Orders	Manual notes and follow-ups	Online order workflow
Credit records	Scattered notebooks or messages	Customer credit ledger
Payments	Difficult to reconcile	UPI option plus visible status
Customer support	Repeated questions	AI-assisted responses

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HOW · THE PROPOSED SYSTEM

One connected system for customers and administrators

Customer-facing features and administrative controls share product, order and customer data; each module addresses an operational gap.

01	02	03	04	05
Browse catalogue	*Select products*	*Place order*	*Payment or credit*	*Support and status updates*



Digital Product Catalogue: searchable products with names, categories, descriptions and pricing fields.



Online Order Management: capture order details, quantities, status and customer communication in one flow.



Customer Credit/Udhaar Management: maintain customer-wise credit entries, payments and outstanding balances.



AI Customer Support: answer common product, order and payment questions using approved business information.



Google Maps Integration: help customers locate the business and plan a visit or delivery interaction.



UPI Payment Facility: provide a convenient digital payment option and support payment-status tracking.



Admin Dashboard: give authorised users a central view of products, orders, customers and summaries.

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SYSTEM DESIGN · ARCHITECTURE

System architecture keeps every action connected

The proposed system separates interface, server logic and database, with optional support, maps and payment services.



Customer: browses, orders, asks questions; selects payment or credit. **React frontend:** catalogue, forms, order screens and dashboard views. **Node.js and Express backend:** validates requests, applies business rules and connects services. **MongoDB:** stores products, orders, customer profiles, payment records and credit-ledger entries. **Connected services:** AI support handles common queries; Google Maps supports location discovery; UPI supports payment initiation.

Order-processing flow

1 *Order placed*

Customer submits request.

2 *Backend validates*

Backend checks request.

3 *Record updated*

Stock and order record updated.

4 *Payment or credit*

UPI or credit path selected.

5 *Admin review*

Authorised user reviews activity.

6 *Status update*

Customer receives next update.

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TECHNOLOGY · APPLICATIONS · OUTCOMES

Technology, applications and expected outcomes

The MERN stack provides a practical foundation for a responsive, maintainable and scalable proposed system.

TECHNOLOGY	PURPOSE IN THE PROPOSED SYSTEM
MongoDB	Store products, orders, customers, payments and credit records.
Express.js	Create backend routes and APIs for system operations.
React.js	Build responsive customer and admin interfaces.
Node.js	Run the server-side application and coordinate services.

APPLICATIONS

Customer ordering and product discovery; business administration; payment and credit tracking.

EXPECTED OUTCOMES

Centralised records; clearer status visibility; easier product discovery; faster responses to common questions.

FUTURE SCOPE

Inventory alerts; delivery tracking; analytics; role-based access; multilingual support.

The proposed system can make construction-material operations more organised, accessible and digitally manageable by connecting customers, administrators, records and services in one workflow.

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